

Econ 8220: Entry Game Problem Set

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Introduction

The main objective of this problem set is to study firm entry decisions and strategic interactions between large discount retailers. In particular, we examine how the presence of a competitor affects a firm's decision to enter a market, and how different modeling assumptions shape what we can learn from the data. This is in the spirit of Bresnahan and Reiss (1991), Jia (2008), and Seim (2006).

Complete this problem set by answering the questions below. When you are finished, submit your completed work as a pdf (or multiple pdfs) with naming convention **'EntryPS_[insert last name]'**. For example, 'EntryPS_Newberry.pdf'. Feel free to have one pdf for your write up and one for your code, labeled `"_code"` and `"_writeup"`, respectively. Feel free to work together but turn in your own write up. You can use any programming language you wish.

Part of the purpose of these problem sets is to help you learn how to write up an academic paper (especially an empirical one), so please include derivations of estimating equations, details of how you estimate the model, intuition about the answers, etc., in your write up.

Set up and Data

Download the data from Jia's (2008) *Econometrica* paper using the following link. I have also added the data from the replication package to eLC. There is a file in there describing the data sets. Upload the file containing the 1997 data into your preferred statistical package.

The data contains information on entry decisions by Wal-Mart and Kmart across U.S. markets (counties), along with a set of market-level characteristics. Throughout this problem set you can use the following control variables: county population (log), sales per capita (log), percentage urban, midwest indicator, south indicator, number of small businesses, and distance to Benton county (Wal-Mart's home county). The latter you may only want to include in Wal-Marts profit function.

The key empirical challenge in this setting is modeling strategic interaction: the entry decision of one firm depends on the entry decision of the other, which generates a simultaneity problem and, in richer models, the possibility of multiple equilibria.

Set up

1. Write down the linear reduced form profit function for retailer i in market m where profit is a heterogeneous function of control variables and a heterogeneous competition effect of the other big retailer. The profit function should contain an error term which is heterogeneous, ϵ_{im} , and can represent the fixed cost of entry.
2. Write down the entry condition for firm i in market m .

Probit Regressions

3. Set up and run **Probit regressions** of entry by Wal-Mart and entry by Kmart on controls (i.e., two separate specifications). First start by setting the competition effects to 0. Put the results into a table.

4. Now include the presence of the competitor into the probit regressions and add the results to the table. What is the biggest weakness of this approach?
5. Now consider an instrumenting strategy that would exploit excluded variables as instruments in the Probit regressions above. In particular, consider the Probit regression of Kmart entry on covariates and Wal-Mart entry. Could **distance from Benton county** serve as a valid instrument for Wal-Mart entry in this regression? Discuss both the relevance and exclusion conditions. Estimate the first stage model (i.e., Wal-Mart entry on controls and distance) and then the second stage model (i.e., K-Mart entry on covariates and the predicted value of Wal-Mart entry from the first stage) and add them both to the table.

Bresnahan-Reiss

6. Now consider a **Bresnahan and Reiss (1991)** analysis of this industry. Set up and estimate an **ordered Probit model** in two specifications, where the dependent variable is (i) the number of large players (Wal-Mart and Kmart), and (ii) the total number of small and large players. Use the same market-level controls as above. Report your estimates in **Table II** and compare with the results in Table I. What are the main limitations of this model?

Incomplete Information

To add firm heterogeneity to the model without confronting the issue of multiple equilibria, suppose Wal-Mart and Kmart play an **entry game of incomplete information**, as in Seim (2006). Under incomplete information, each firm forms a belief $P_{-i,m}^*$ about the entry probability of its rival and best-responds to that belief. The equilibrium entry probabilities $P_{i,m}^*$ are implicitly defined by the fixed point condition:

$$P_{i,m}^*(\beta, \Delta) = \frac{\exp(x_{im}\beta_i + \Delta_i P_{-i,m}^*)}{1 + \exp(x_{im}\beta_i + \Delta_i P_{-i,m}^*)}, \quad i \in \{M, K\}.$$

7. Set up an estimate the model using the following **nested fixed-point (NFP) maximum likelihood estimator**:

$$\max_{\beta, \Delta} \sum_i \sum_m [y_{im} \ln P_{i,m}^*(\beta, \Delta) + (1 - y_{im}) \ln(1 - P_{i,m}^*(\beta, \Delta))] \quad (1)$$

$$\text{s.t. } P_{i,m}^*(\beta, \Delta) = \frac{\exp(x_{im}\beta_i + \Delta_i P_{-i,m}^*)}{1 + \exp(x_{im}\beta_i + \Delta_i P_{-i,m}^*)}, \quad i \in \{M, K\}. \quad (2)$$

Note that the likelihood can be written down as the separate likelihood of entry for each firm in market m like the above because those probabilities are conditionally independent.

The NFP estimator works by nesting an **inner loop** that solves for the equilibrium probabilities $\mathbf{P}_m^* = \{P_{W,m}^*, P_{K,m}^*\}$ via a contraction mapping (iterating best responses until convergence) inside an **outer loop** that maximizes the log-likelihood over the structural parameters $\theta = \{\beta_W, \Delta_W, \beta_K, \Delta_K\}$. For the inner loop, initialize at $P^{(0)} = \{0.5, 0.5\}$ and iterate until the log-difference stopping rule $\max_i \left| \ln(P_i^{(t)} / P_i^{(t-1)}) \right| \leq \tau$ is satisfied (e.g., $\tau = 10^{-6}$). Report your estimates and compare them to the results in Tables I and II.

References

1. Bresnahan, Timothy F., and Peter C. Reiss. "Entry and competition in concentrated markets." *Journal of Political Economy* 99.5 (1991): 977–1009.

2. Bajari, Patrick, et al. “Estimating static models of strategic interactions.” *Journal of Business & Economic Statistics* (2012).
3. Jia, Panle. “What happens when Wal-Mart comes to town: An empirical analysis of the discount retailing industry.” *Econometrica* 76.6 (2008): 1263–1316.
4. Seim, Katja. “An empirical model of firm entry with endogenous product-type choices.” *The RAND Journal of Economics* 37.3 (2006): 619–640.
5. Su, Che-Lin. “Estimating discrete-choice games of incomplete information: Simple static examples.” *Quantitative Marketing and Economics* 12.2 (2014): 167–207.